

DMX Desk Emulator

Fixture Definition Reference

A complete guide to writing and editing fixture JSON files

Fixture definitions tell the desk how to display and control a DMX fixture — what channels it has, their labels, ranges, and behaviour. Each fixture type is defined in a **.json** file in the **fixtures/** folder. The filename (without .json) is used as the **type** field in patch.json.

1. File Structure

A fixture definition file is a single JSON object with one required key: **channels**. It may also have a top-level **lockable** flag.

```
{
  "channels": [ ... ],
  "lockable": false
}
```

Field	Type		Description
channels	array	Required	List of channel definition objects, one per DMX channel. Order matches DMX address offset (first entry = base address).
lockable	boolean	Optional	If true, lock buttons appear on the fixture faceplate. Locked fixtures are excluded from scene recalls. Default: false. Use for house lights, special effects etc.

2. Channel Fields

Each entry in the channels array defines one DMX channel. Channels are assigned DMX addresses sequentially starting from the fixture's base address.

Field	Type		Description
label	string	Required	Short display name shown on the fader. Max ~8 characters for best layout. Special labels R, G, B, W, A, UV trigger colour swatch display.
master	boolean	Required	Exactly one channel per fixture should be true. This channel is shown as the large master fader. All other channels are shown as smaller sub-channel faders.
default	number	Required	Default value sent to DMX when the fixture is first loaded and for hidden channels. Must be within the channel's range.
range	varies	Required	Defines the fader range. Either a numeric array [min, max] or a named dict {"lo-hi": "name"}. See Section 3.
unit	string	Required	How the value is displayed and scaled. One of: %, raw, named. See Section 4.
show	boolean	Required	If true, the channel appears as a visible fader. If false, the channel is hidden — its default value is sent to DMX but the operator cannot adjust it from the desk.

jump	boolean	Optional	If true, this channel snaps instantly during scene fades rather than interpolating. Use for mode/program/strobe channels where fading through intermediate values is undesirable. Snap direction is automatic: snaps before fade-in (when master increases) or after fade-out (when master goes to zero). Default: false.
role	string	Optional	Assigns a special role to this channel. Currently supported: "pan" and "tilt". If both pan and tilt roles are present, an XY joystick pad appears on the faceplate alongside the colour faders. Default: none.

3. The range Field

3.1 Numeric Range

A two-element array [**min**, **max**] defines a continuous numeric fader. The fader moves between these values. DMX output is scaled proportionally to the 0–255 DMX range.

```
"range": [0, 255]    // raw 0-255
"range": [0, 100]    // percentage 0-100%
```

Use **[0, 100]** with **"unit": "%"** for intensity channels — the desk displays values as percentages and scales to 0–255 DMX internally. Use **[0, 255]** with **"unit": "raw"** for channels where you want direct DMX value control.

3.2 Named Range

A JSON object maps value ranges to display names. Each key is either a single value or a range in the form **"lo-hi"**. The fader snaps between the defined zones and the label updates to show the current zone name.

```
"range": {
  "0-31": "LED Off",
  "32-63": "LED On",
  "64-95": "Strobe",
  "96-127": "LED On",
  "128-255": "Pulse"
}
```

Named ranges are ideal for strobe, colour wheel, gobo wheel, program select, and mode channels — anywhere the DMX value selects a discrete function rather than a continuous level. Always combine with **"unit": "named"** and consider adding **"jump": true** (see Section 5).

Note: The key is the start value of the zone. The desk determines zone boundaries from consecutive keys. The last key's zone extends to 255. Keys must be quoted strings even though they represent numbers.

4. The unit Field

Value	Meaning
%	Percentage display. Range should be [0, 100]. Value is shown as 0%–100% on the fader. DMX output is scaled: 100% = DMX 255, 50% = DMX 127 etc. Use for intensity, dimmer, and colour mixing channels.

raw	Direct DMX value display. Range should be [0, 255]. Value shown as 0–255. No scaling — fader value = DMX value. Use when you need precise DMX control or when the fixture manual specifies exact DMX values.
named	Named selector display. Range must be a named dict (see 3.2). Fader becomes a selector that snaps between named zones. The zone name is shown instead of a number. Use for mode, program, strobe, colour wheel, gobo channels.

5. Jump Channels ("jump": true)

When a scene is recalled with a fade time, most channels interpolate smoothly from their current value to the target value. Jump channels bypass this and snap instantly — like professional console mark/unmark behaviour.

5.1 Snap at Start (fade-in)

When the fixture's master channel is increasing (or going from zero), jump channels snap to their target value **at the beginning** of the fade, before intensity rises. The fixture arrives at full brightness already in the correct mode, colour, or gobo position.

5.2 Snap at End (fade-out)

When the fixture's master channel is going to zero, jump channels snap **after** the fade completes. The audience never sees the channel fading through intermediate values — the change happens silently in the dark.

5.3 When to use jump

Add **"jump": true** to channels where fading through intermediate values would be wrong or ugly:

Value	Meaning
Strobe	Fading from 0 to 64 would briefly strobe — snap instead.
Colour wheel	Fading through intermediate positions shows wrong colours.
Gobo wheel	Intermediate positions show split gobos — always snap.
Program select	Stepping through programs looks unintentional.
Mode channels	Any channel that selects a mode rather than a level.
Prism	Intermediate values may cause unexpected effects.
Reset	Should never interpolate — always jump.
Dimming curves	Changing curve mid-fade produces unpredictable results.

6. The role Field

The role field assigns special behaviour to a channel beyond its normal fader display.

Value	Meaning
"role": "pan"	Pan channel for a single-head fixture. When both pan and tilt roles are present, an XY joystick pad appears. Dragging sets both simultaneously. The fader remains visible for fine control.

"role": "tilt"	Tilt channel for a single-head fixture. Works with pan role to enable the XY pad. Default value 128 (centre) recommended.
"role": "pan1" / "pan2"...	Numbered pan role for multi-head fixtures (scanner bars etc). When numbered pairs are found (pan1/tilt1, pan2/tilt2 etc), a single XY pad appears with a Head dropdown selector. Selecting a head switches the pad to control that head. All heads faders remain independently accessible.
"role": "tilt1" / "tilt2"...	Numbered tilt role matching the corresponding numbered pan. The desk detects all complete pairs automatically. A 4-head bar would use pan1/tilt1 through pan4/tilt4.

Note: The XY pad appears in the same row as colour faders (R/G/B/W etc) to keep the fixture widget compact. If there are no colour channels, the pad gets its own row. Pan fine and tilt fine channels (16-bit) should be set to "show": false to keep the faceplate clean — the XY pad provides sufficient positional control for most purposes.

7. The lockable Flag

Set at the top level of the fixture definition (not inside a channel). When true, a lock button appears on the fader for each channel.

```
{
  "channels": [ ... ],
  "lockable": true
}
```

Locking a channel prevents it from being changed by scene recalls. The channel stays at its current value regardless of what the recalled scene specifies. An **UNLOCK ALL** button in the footer clears all locks.

Use lockable for fixtures that must not be affected by show cues — house lights, work lights, and special effects cannons are the primary use cases. A dedicated **housetlights.json** fixture type is provided which is identical to **dimmer.json** but with **lockable: true**.

8. Complete Examples

8.1 Simple Dimmer

```
{
  "channels": [
    {
      "label": "Dimmer", "master": true, "default": 0,
      "range": [0, 100], "unit": "%", "show": true
    }
  ]
}
```

8.2 RGBW LED Fixture

```
{
  "channels": [
    {
      "label": "Intensity", "master": true, "default": 0,
      "range": [0, 100], "unit": "%", "show": true,
    },
    {
      "label": "R", "master": false, "default": 0,
      "range": [0, 100], "unit": "%", "show": true,
    },
    {
      "label": "G", "master": false, "default": 0,
      "range": [0, 100], "unit": "%", "show": true,
    },
    {
      "label": "B", "master": false, "default": 0,
      "range": [0, 100], "unit": "%", "show": true,
    },
    {
      "label": "W", "master": false, "default": 0,
      "range": [0, 100], "unit": "%", "show": true,
    }
  ]
}
```

```

    {"label": "B",          "master": false, "default": 0,
     "range": [0, 100], "unit": "%", "show": true},
    {"label": "W",          "master": false, "default": 255,
     "range": [0, 100], "unit": "%", "show": true}
  ]
}

```

8.3 Fixture with Strobe and Program (jump channels)

```

{
  "channels": [
    {"label": "R",          "master": false, "default": 0,
     "range": [0, 100], "unit": "%", "show": true},
    {"label": "G",          "master": false, "default": 0,
     "range": [0, 100], "unit": "%", "show": true},
    {"label": "B",          "master": false, "default": 0,
     "range": [0, 100], "unit": "%", "show": true},
    {"label": "Strobe",     "master": false, "default": 32,
     "range": {"0-31": "Off", "32-63": "On", "64-255": "Strobe"},
     "unit": "named", "show": true, "jump": true},
    {"label": "Program",    "master": false, "default": 0,
     "range": {"0-10": "Off", "11-50": "Prog 1", "51-100": "Prog 2",
              "101-255": "Sound Active"},
     "unit": "named", "show": true, "jump": true},
    {"label": "Master",     "master": true, "default": 0,
     "range": [0, 100], "unit": "%", "show": true}
  ]
}

```

8.4 Moving Head with Pan/Tilt XY Pad

```

{
  "channels": [
    {"label": "Pan",          "master": false, "default": 128,
     "range": [0, 255], "unit": "raw", "show": true, "role": "pan"},
    {"label": "Pan Fine",     "master": false, "default": 0,
     "range": [0, 255], "unit": "raw", "show": false},
    {"label": "Tilt",         "master": false, "default": 128,
     "range": [0, 255], "unit": "raw", "show": true, "role": "tilt"},
    {"label": "Tilt Fine",    "master": false, "default": 0,
     "range": [0, 255], "unit": "raw", "show": false},
    {"label": "Dimmer",       "master": true, "default": 0,
     "range": [0, 255], "unit": "raw", "show": true},
    {"label": "Colour",       "master": false, "default": 0,
     "range": {"0-9": "White", "10-19": "Red", "20-29": "Green",
              "30-39": "Blue", "40-49": "Yellow"},
     "unit": "named", "show": true, "jump": true},
    {"label": "Gobo",         "master": false, "default": 0,
     "range": {"0-9": "Open", "10-19": "Gobo 1", "20-29": "Gobo 2"},
     "unit": "named", "show": true, "jump": true}
  ]
}

```

8.5 Multi-Head Scanner Bar (numbered roles)

```

{
  "channels": [
    {"label": "Blackout", "master": true, "default": 0,

```

```

    "range": {"0-127": "Blackout", "128-255": "Full On"},
    "unit": "named", "show": true, "jump": true},
  {"label": "1 Pan", "master": false, "default": 128,
   "range": [0, 255], "unit": "raw", "show": true, "role": "pan1"},
  {"label": "1 Tilt", "master": false, "default": 128,
   "range": [0, 255], "unit": "raw", "show": true, "role": "tilt1"},
  {"label": "2 Pan", "master": false, "default": 128,
   "range": [0, 255], "unit": "raw", "show": true, "role": "pan2"},
  {"label": "2 Tilt", "master": false, "default": 128,
   "range": [0, 255], "unit": "raw", "show": true, "role": "tilt2"}
]
}

```

8.6 Lockable House Lights

```

{
  "channels": [
    {"label": "Dimmer", "master": true, "default": 0,
     "range": [0, 100], "unit": "%", "show": true}
  ],
  "lockable": true
}

```

9. Quick Reference

Colour swatch automatic activation

A colour swatch appears automatically when the fixture has channels with labels from this list: R, G, B, W, A, UV. No special configuration needed.

DMX scaling rules

Value	Meaning
unit: "%", range: [0,100]	Fader 0–100 → DMX 0–255 (linear scale)
unit: "raw", range: [0,255]	Fader 0–255 → DMX 0–255 (no scaling)
unit: "named"	Fader value sent directly to DMX with no scaling

Hidden channel defaults

Hidden channels ("show": false) send their default value to DMX immediately when the fixture is loaded. This is the correct way to handle channels like Pan Fine / Tilt Fine, reserved channels, or mode-select channels that should always be at a fixed value.

File naming

The filename (without .json) is used as the fixture type in patch.json. Use lowercase with underscores: **my_fixture.json** → **"type": "my_fixture"**. Avoid spaces and special characters in filenames.

Loading and caching

Fixture definitions are cached in memory when first loaded. After editing a .json file, use **Save & Reload** in the patch editor to pick up the changes. The cache clears automatically on reload.